



Assignments

Recursion – 1

Q1. Write a program to calculate the sum of odd numbers between a and b (both inclusive) using recursion.

Solution :

```
#include <bits/stdc++.h>
using namespace std;
int findSum(int curr, int lastNumber) {
    if(curr > lastNumber) return 0;
    if(curr % 2 == 0) return findSum(curr+1, lastNumber);
    return curr + findSum(curr+2, lastNumber);
}
int main() {
    int a, b;
    cin >> a >> b;
    cout << findSum(a, b) << endl;
    return 0;
}
```

Q2. Calculate the number of ways in which a person can climb n stairs if he can take exactly 1, 2 or 3 steps at each level.

Solution :

```
#include <bits/stdc++.h>
using namespace std;
int findNumberOfWays(int n) {
    if(n < 0) return 0;
    if(n == 0) return 1;
    return findNumberOfWays(n-1) + findNumberOfWays(n-2) +
    findNumberOfWays(n-3);
}
int main() {
    int n;
    cin >> n;
    cout << findNumberOfWays(n) << endl;
    return 0;
}
```

Q3. Given a positive integer, return true if it is a power of 2.

Solution :

```
#include <bits/stdc++.h>
using namespace std;
bool isPowerOfTwo(int n) {
    if(n == 1) {
        return true;
    }
}
```

```
if(n % 2 == 0) {  
    return isPowerOfTwo(n / 2);  
}  
return false;  
}  
int main() {  
    int n;  
    cin >> n;  
    if(isPowerOfTwo(n)) {  
        cout << "Yes" << endl;  
    } else {  
        cout << "No" << endl;  
    }  
}
```





**THANK
YOU !**

